

MATERIAL SAFETY DATA SHEET (MSDS)

Standard GHS Compliance Document for Laboratory Safety

SECTION 1: CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Chemical Name	Hydrochloric Acid, 37%
Chemical Formula	HCl
CAS Number	7647-01-0
Document Purpose	Academic RAG Validation Dataset

SECTION 2: HAZARDS IDENTIFICATION

 Corrosive, Acute Toxicity (Oral/Inhalation), Skin Corrosion (Category 1B), Serious Eye Damage (Category 1).

SECTION 4: FIRST AID MEASURES

- Inhalation:** Move person into fresh air. If not breathing, give artificial respiration. Consult a physician immediately.
- Skin Contact:** Take off contaminated clothing immediately. Wash off with soap and plenty of water for at least 15 minutes.
- Eye Contact:** Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician. Continue rinsing eyes during transport to hospital.
- Ingestion:** Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Rinse mouth with water. Consult a physician.

SECTION 6: ACCIDENTAL RELEASE MEASURES

- Personal Precautions:** Use personal protective equipment. Avoid breathing vapors, mist or gas. Ensure adequate ventilation. Evacuate personnel to safe areas.
- Environmental Precautions:** Prevent further leakage or spillage if safe to do so. Do not let product enter drains.
- Clean-up Methods:** Soak up with inert absorbent material (e.g., sand, silica gel, acid binder). Neutralize carefully with basic materials like lime or sodium carbonate. Clean surface thoroughly.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Property	Value
Appearance	Clear, colorless liquid
Odor	Pungent, irritating
pH	< 1 (strongly acidic)
Boiling Point	108.5 °C (227 °F)
Melting Point	-30 °C (-22 °F)
Density	1.19 g/cm³

SECTION 10: STABILITY AND REACTIVITY

Stable under recommended storage conditions. Reacts violently with strong bases, strong oxidizing agents, metals (yields hydrogen gas which is highly flammable).

SECTION 16: OTHER INFORMATION

This document is synthetically generated for testing Retrieval-Augmented Generation (RAG) system pipelines within clinical biochemistry and safety management training workflows.